



Property Valuation — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 04-April-2026

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1 Test Results — Property Valuation (MKM-PV-001)

Tests run on **04 April 2026 at 14:48:33**.

Table 1: Test Results Summary — Property Valuation (MKM-PV-001)

Total Tests	55
Passed	55
Failed	0
Skipped	0
Duration	0.15s

Table 2: Detailed Test Results — Property Valuation

Test Description	Acceptance Criteria	Result	Time
Line 541: non-dict schema → returns {}.	result == {}	Pass	0.000s
Line 544-545: 'type'/'options'/'description' fields are skipped.	"type" not in result; "options" not in result	Pass	0.000s
Calculate insurance premium flood risk high	premium > 0	Pass	0.001s
Line 214: area=None → computes it first.	isinstance(premium, float); premium > 0	Pass	0.001s
Line 211: property_value=None → computes it first.	isinstance(premium, float); premium > 0	Pass	0.001s
Calculate monthly rent all types	800 <= rent <= 15.000	Pass	0.000s
Lines 165-166: area=None → computes it first.	isinstance(rent, float); rent > 0	Pass	0.000s
Lines 162-163: property_value=None → computes it first.	isinstance(rent, float); rent > 0	Pass	0.001s
Calculate monthly rent positive	isinstance(rent, float); 800 <= rent <= 15.000	Pass	0.000s
Calculate property area all types	area > 0	Pass	0.000s
Lines 128-129: invalid date format → random years_ago.	isinstance(price, float)	Pass	0.000s
Lines 120-121: property_value=None → computes it first.	isinstance(price, float); price > 0	Pass	0.001s
Calculate sale price positive	isinstance(price, float); 50.000 <= price <= 8.000.000	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Lines 135-137: sale_date 1-2 years ago.	<code>isinstance(price, float)</code>	Pass	0.000s
Lines 139-140: sale_date > 2 years ago.	<code>isinstance(price, float)</code>	Pass	0.000s
Lines 124-128: sale_date within 1 year.	<code>isinstance(price, float)</code>	Pass	0.000s
Line 60-61: datetime → isoformat string.	<code>"2024-01-01" in result</code>	Pass	0.000s
Lines 64-65: np.float64 → float.	<code>"3.14" in result</code>	Pass	0.000s
Lines 62-63: np.integer → int.	<code>result == "42"</code>	Pass	0.000s
Lines 66-67: np.ndarray → list.	<code>result == [1, 2, 3]</code>	Pass	0.000s
Line 68: super().default() raises TypeError for unknown type.	—	Pass	0.000s
Lines 510-517: each fallback location has lat/lon/name/elevation.	<code>"lat" in loc; "lon" in loc (+2 more)</code>	Pass	0.001s
Lines 499-518: fallback generates right number of locations.	<code>len(locs) == 5</code>	Pass	0.000s
Lines 505-506: uses params.get_elevation if available.	<code>all(loc["elevation"] == 10.0 for loc in locs)</code>	Pass	0.000s
Lines 608-612: generate_properties returns result dict.	<code>isinstance(result, dict); "data" in result (+1 more)</code>	Pass	0.018s
Lines 610-612: generate_properties creates output file.	<code>result["file_path"].exists()</code>	Pass	0.004s
Calculate mortgage financials returns dict	<code>isinstance(result, dict); "loan_amount" in result (+2 more)</code>	Pass	0.000s
Generate financial data keys	<code>"mortgage_id" in data; "loan_amount" in data (+1 more)</code>	Pass	0.000s
Interest rate range	<code>0.01 <= data["interest_rate"] <= 0.15</code>	Pass	0.000s
Loan amount less than value	<code>data["loan_amount"] <= info["property_value"] * 1.05</code>	Pass	0.000s
Mortgage id format	<code>re.match(r"~MORT-[a-f0-9]{8}\$", data["mortgage_id"])</code>	Pass	0.000s
Quality consistency check	<code>isinstance(corrected, dict)</code>	Pass	0.000s
Calculate insurance premium positive	<code>isinstance(premium, (int, float)); premium > 0</code>	Pass	0.001s
Calculate property area positive	<code>isinstance(area, (int, float)); area > 0</code>	Pass	0.000s
Calculate property value in range	<code>50_000 <= value <= 10_000_000</code>	Pass	0.004s
Generate bedrooms detached	<code>isinstance(beds, int); 1 <= beds <= 7</code>	Pass	0.000s
Generate bedrooms flat	<code>isinstance(beds, int); 1 <= beds <= 4</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Generate postcode format	<code>isinstance(postcode, str); len(postcode) >= 5</code>	Pass	0.000s
Generate field value decimal	<code>isinstance(val, (int, float))</code>	Pass	0.001s
Generate field value text	<code>isinstance(val, str)</code>	Pass	0.001s
Coordinates in thames range	<code>51.0 <= loc["lat"] <= 52.0; -1.0 <= loc["lon"] <= 1.0</code>	Pass	0.018s
Processing stats	<code>stats["successful_properties"] == 5; stats["failed_properties"] == 0</code>	Pass	0.012s
Output file exists	<code>result["file_path"].exists()</code>	Pass	0.009s
Output file is valid json	<code>isinstance(data, dict)</code>	Pass	0.014s
Generate correct count	<code>len(result["data"]["properties"]) == 5</code>	Pass	0.008s
Generate returns expected keys	<code>"data" in result; "file_path" in result (+1 more)</code>	Pass	0.015s
Property id format	<code>re.match(r"~PROP-[a-f0-9]{8}\$", pid)</code>	Pass	0.014s
Property ids unique	<code>len(ids) == len(set(ids))</code>	Pass	0.022s
Construction year in range	<code>1600 <= year <= 2026</code>	Pass	0.000s
Property id format	<code>re.match(r"~PROP-[a-f0-9]{8}\$", meta["property_id"])</code>	Pass	0.000s
Returns dict with required keys	<code>"property_id" in meta; "property_type" in meta</code>	Pass	0.001s
Lines 73-75: generate_grid_reference returns a non-empty string.	<code>isinstance(ref, str); len(ref) > 4</code>	Pass	0.000s
Lines 50-52: generate_owner_name returns 'Firstname Lastname'.	<code>len(parts) == 2; all(p[0].isupper() for p in parts)</code>	Pass	0.000s
Lines 43-45: generate_past_date returns YYYY-MM-DD string.	<code>isinstance(result, str); len(parts) == 3</code>	Pass	0.000s
Custom days_range is respected (result is a past date).	<code>dt < datetime.now()</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (28 tests)	D. Kelly
08-Mar-2026	1.1	25 test(s) added; 28 test(s) removed (total: 25)	D. Kelly
08-Mar-2026	1.2	4 test(s) added (total: 29)	D. Kelly
09-Mar-2026	1.3	12 test(s) added (total: 41)	D. Kelly
04-Apr-2026	1.4	14 test(s) added (total: 55)	D. Kelly